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Artificial intelligence for social good

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Abstract

Artificial Intelligence (AI) presents transformative opportunities but also significant challenges. Training large AI models consumes vast energy, driving carbon emissions, which can be mitigated through renewable energy and sustainable practices. International cooperation is essential to address regulatory gaps and ensure ethical, equitable AI use. To prevent deepening social inequalities, AI must prioritize inclusive design, diverse community engagement, and fairness-focus regulations. If developed responsibly, AI will contribute significantly to urgent global challenges ranging from climate change, healthcare, and education toward a greener, more equal future.

Artificial intelligence for social good

Introduction:

AI finds some of its most impactful applications in social problems. The examination of these elements raises ethical questions necessary for responsible AI development and use. I'm thinking about environmental issues around A.I. model training and international cooperation necessary to respond A.I fueled cultural change and social inequality problems, or more generally answering the question if A.I. technologies are able to address climate change issues. In relation to the worrying environmental impact, the amount of energy required to train such models renders any purpose moot:

It has been established that massive A.I. models contribute to carbon emissions, with hundreds of billions of dollars worth of A.I. models being trained each year, which when scaled would contribute significantly to A.I. carbon emissions. A study conducted by Strubell noted that A.I training could contribute the same amount of carbon emissions as 5 cars would in their lifetime.

Due to massive data requirements, A.I requires colossal servers and cooling systems which further contributes to energy consumption, as noted by Jones. A complete switch to using renewable energy for these operations could offset the carbon footprint which would significantly reduce climate implications CO2 infusion would create. In addressing the climate concern, massive A.I. models should be avoided.

1.2 Ethics

The influence of AI on the environment brings a number of ethical aspects to consider.

Responsibility for Sustainability: Organizations should not simply strive for new advancements by all means but rather ensure that new AI efforts are sustainable for future generations (Binns, 2018). AI Tax on Carbon: Just like for businesses involving carbon offsets, governments should enforce emissions reporting thresholds to accompany the development of AI systems (Vincent, 2020).

Lifecycle Assessment: To fully assess the AI Model's technology and its possible effects on the environment in its entirety, it is necessary to appreciate the lifecycle of AI models which ranges from development to deployment (Patterson et al. 2021).

2. International Cooperation and Regulation / Governance Systems

2.1 Need for Cooperation

AI in its functions is universal, therefore cross boundaries cooperation is inevitable.

AI Technologies are Core Technologies: The ethical concerns that revolve around the AI application are out of boundaries, thus having a unified approach and creating relevant governance frameworks are necessary (Jobin et al., 2019).

AI Responsibility: Unity in ethical principles can mitigate risks interference in AI responsible creation and use among other development (OECD, 2021).

2.2 Hurdles to Cooperation

Nonetheless, it can be observed that the governance of AI at an international level is challenging due to a number of aspects:

Regulatory Divergence: Different countries have different regulations and ethical standards, thus preventing the establishment of an internationally integrated AI governance mechanism (Gonzalez et al., 2020).

Power Dynamics: As a result, wealthier countries are the major actors and dominators of AI research and Technological development on the international scene, which creates a situation where they establish regulatory biases that also extend globalization inequalities (Crawford, 2021).

2.3 Potential Solutions

Some of the techniques that we can use in order to address these challenges include the following:

International bodies: This could be responsible and international organizations about AI Ethics (OECD), UE global Data Governance, AI Policy and AI Governance can rise accountability and make fairness not an the idea that is only very good (UNESCO, 2021) to international institutions.

Harmonization and Cooperation: At the same time international regulatory Consolidation and Cooperation: international campaign to support and protect more just norms and the interests of multiple regulatory paradigms can foster justice (European Commission, 2020).

3. AI and Social Inequalities

3.1 Contributions to Inequalities

AI both enable and harm social equity:

AI systems developed using biased datasets are capable of propagating the existing disparities, which can lead to discriminatory outcomes in diverse sectors (O'Neil, 2016).

Widening Socio-economic Gaps: Unequal access to generative AI technologies can exacerbate socio-economic disparities, further marginalizing already disadvantaged communities (Noble, 2018).

3.2 How to Advocate in Favor of Social Justice

We call for the following strategies to ensure that AI works for social equity:

Inclusive design practices: Incorporating diverse datasets and methodologies in developing AI systems help minimize bias and improve fairness (Buolamwini & Gebru, 2018).

Engagement with Marginalized Communities : Involving underrepresented populations in the creation and execution of AI solutions helps ensure that their needs and voices are heard (Binns, 2018).

Regulatory Oversight to Ensure Fairness: Regulation aimed at AI fairness and accountability can reduce social injustice and promote equity (European Commission, 2020).

4. AI Applied to Solve Global Challenges

4.1 Climate Change

AI has enormous potential to address climate change in a myriad of innovative ways:

Predictive Analytics: AI can process massive datasets to forecast climate trends, aiding policy makers in making informed decisions (Rolnick et al., 2019).

Facilitating Resource Management Efficiency : AI technologies could make resource efficiency more standardized, helping reduce waste and supporting sustainability in numerous fields (Kumar et al., 2020). **Improving Disaster Response:** AI can enhance disaster response by predicting the occurrence of natural disasters and optimizing logistics for relief efforts (Zhou et al., 2021).

4.2 Other Global Problems

AI can also help combat other major global problems beyond climate change:

Healthcare Enhancement: AI has the potential to improve healthcare while enabling better delivery and access especially in the currently underserved regions while making processes quicker and personal (Esteva et al., 2019).

Educational Equity: AI-powered personalized learning experiences can bridge educational gaps and ensure equitable access to quality education (Luckin et al., 2016).

Conclusion

It is well established that AI can also be leveraged for the social good, but it brings along a lot of challenges that need to be tackled effectively. We should embrace sustainability, elevate international cooperation and amplify social justice to leverage AI to solve some of the world's biggest problems. A balance is critical to the use of speech technologies in a manner that benefits both society and the environment.

Recommendations

Sustainable Practice: The organizations should prefer using renewable energy resources for training AI and reduce their carbon emissions.

Institute Global Governance: Nations need to work together to establish global institutions that govern AI ethical practices and collaboratively develop ethical norms.

Inclusion: Ensure diverse perspectives are part of AI development to enable equitable outcomes.

Contributors to Global Issues for AI. Leverage AI to tackle global challenges such as climate change, health care disparities, and education inequities.

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